

Foundry

# Introduction

- Casting Principle - property of liquid to take up the shape of the vessel.
- Process of Metal casting- pouring of metal into the mold (cavity formed in some molding material- eg: sand)
- Mold cavity resembles shape and size of product to be made.
- Molten metal poured- allowed to freeze- takes shape of mold cavity- casting.

# Foundry

## 5.2 A FOUNDRY SHOP

A **foundry shop** is that shop where molding (making of mold), metal melting and casting the molds and their related processes, such as fettling and cleaning of castings, are conducted. To function properly, the foundry shop is divided into the following departments.

- (i) **Molding department**, where operations related to making of molds and their baking are carried out.
- (ii) **Core department**, where operations related to making and baking of cores are carried out.
- (iii) **Metal melting department**, where melting of metal for casting purposes is carried out in a cupola or other furnaces.
- (iv) **Casting treating or fettling department**, where cleaning of castings and removing of gates, risers or fins from castings are done.
- (v) **Quality control department** is responsible for maintaining a particular quality standard of foundry products (casting) through proper inspection at various stages of production, right from material procurement to the inspection and testing of the castings.

### 5.3 TYPES OF FOUNDRIES

Foundries are usually categorized according to the type of metals or alloys cast in a particular foundry, for example,

- (a) **Cast iron foundries** involve casting of different types of cast irons.
- (b) **Malleable iron foundries** are devoted to the casting of malleable cast iron products.
- (c) **Steel foundries** involve casting of carbon steels and alloy steel products.
- (d) **Non-ferrous foundries** involve casting of non-ferrous metals and alloys.

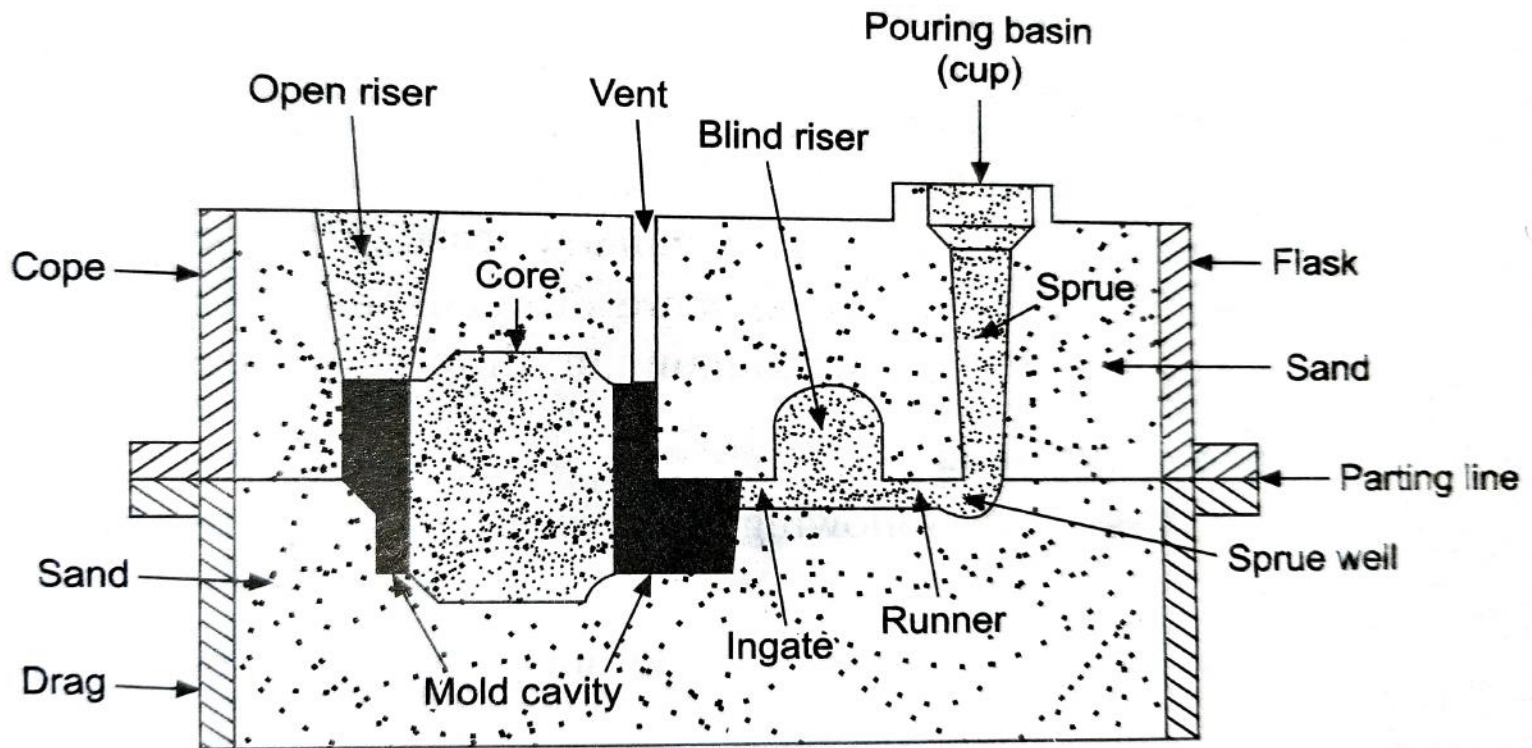
# Advantages of Casting

- (i) It is possible to cast the products with intricate external shapes and complex internal profiles making use of the ability of molten metal to fill completely the cavities in a mold.
- (ii) Casting may be advantageous when a large number of similar parts are produced, thus covering up the cost incurred on patterns.
- (iii) Casting permits the designer to place bulk of metal in a product where it is most needed and remove it from the place where it is in excess, for example, as in case of cast crankshafts and machine tool components.
- (iv) Casting is a suitable process for making very heavy components of structures or machine tool frames. Casting is easier and cheaper in such cases.
- (v) Parts requiring the property of damping of sound and mechanical vibrations are best made by casting, grey cast iron castings usually.
- (vi) Components made of refractory metals or highly creep resistant metals and alloys (for the parts of gas turbine) which are difficult to forge or machine are easily cast to close dimensional tolerances.
- (vii) Casting minimizes anisotropic qualities (or directional qualities found in most wrought or rolled and forged products) which render the rolled component poor in fatigue and impact in the transverse direction of rolling.
- (viii) Casting of precious metals such as gold and silver does not suffer from the problem of loss of material in the form of chips, flash or scrap.

### 5.5.1 Molding

A mold is a void or cavity created in a compact sand mass with the help of a **pattern**, which is the near replica of the casting to be made. This cavity is filled with molten metal, which, on solidification, results into a **casting**. The pattern nearly resembles with the shape and size of the casting to be made. The process of creating the cavity or making of mold (in the sand) is termed **molding**. It consists of compacting molding sand around a pattern (made of wood, plastic or metal) enclosed in a mold box or flask. **Molding-related activities** include: pattern making, preparing molding sand mixture, core making, baking of mold and core and preserving molds in case of dry sand molds and metallic molds. Essential features of a sand mold and few terms related to the mold are explained in the following in reference to Fig. 5.1.





**Fig. 5.1** Illustrating salient features of a sand mold.

**Pouring basin** or **pouring cup** is an enlarged cup-shaped cavity made at the top end of a vertical sprue to help easy feeding of molten metal into the mold from the ladle during the process of casting.

**Sprue** is a vertical entry for the molten metal to flow downward to runners and finally to the mold cavity.

**Runners** are the channels to carry molten metal from the sprue to the mold cavity.

**Risers** are specifically designed to store some additional metal during pouring so that when the casting shrinks during its solidification, the metal from the risers may flow to the casting to compensate for its shrinkage. Risers are designed to keep the metal in them in molten state until the casting is fully solidified. This way, the risers are the last to solidify. Risers also help in escaping out (from the mold) of the entrapped gases and slag which may be dissolved (or in suspension) with the molten metal. Risers may be **open risers** or **blind risers**. Open risers indicate the complete filling of the mold by overflowing the molten metal from the top of the risers during casting [Fig. 5.2(b)].



**Core** is the insert (made of sand) placed in the mold cavity to form hollow regions within the casting or to define the interior surfaces of casting. Cores are also used on the outside of casting to form features such as deep external pockets or contours. Cores are made of sand and some binding material and are usually baked for strength before use. Metal cores are used with permanent metallic molds.

**Vents** are passages to carry off air and gases produced in the mold cavity when the molten metal comes in contact with the mold walls and the cores, since both are usually made from sands having additives which, on burning, generate lots of gases.

**Gates** or **ingates** are the entry ends of the runners connecting the mold cavity or risers with the main runners.

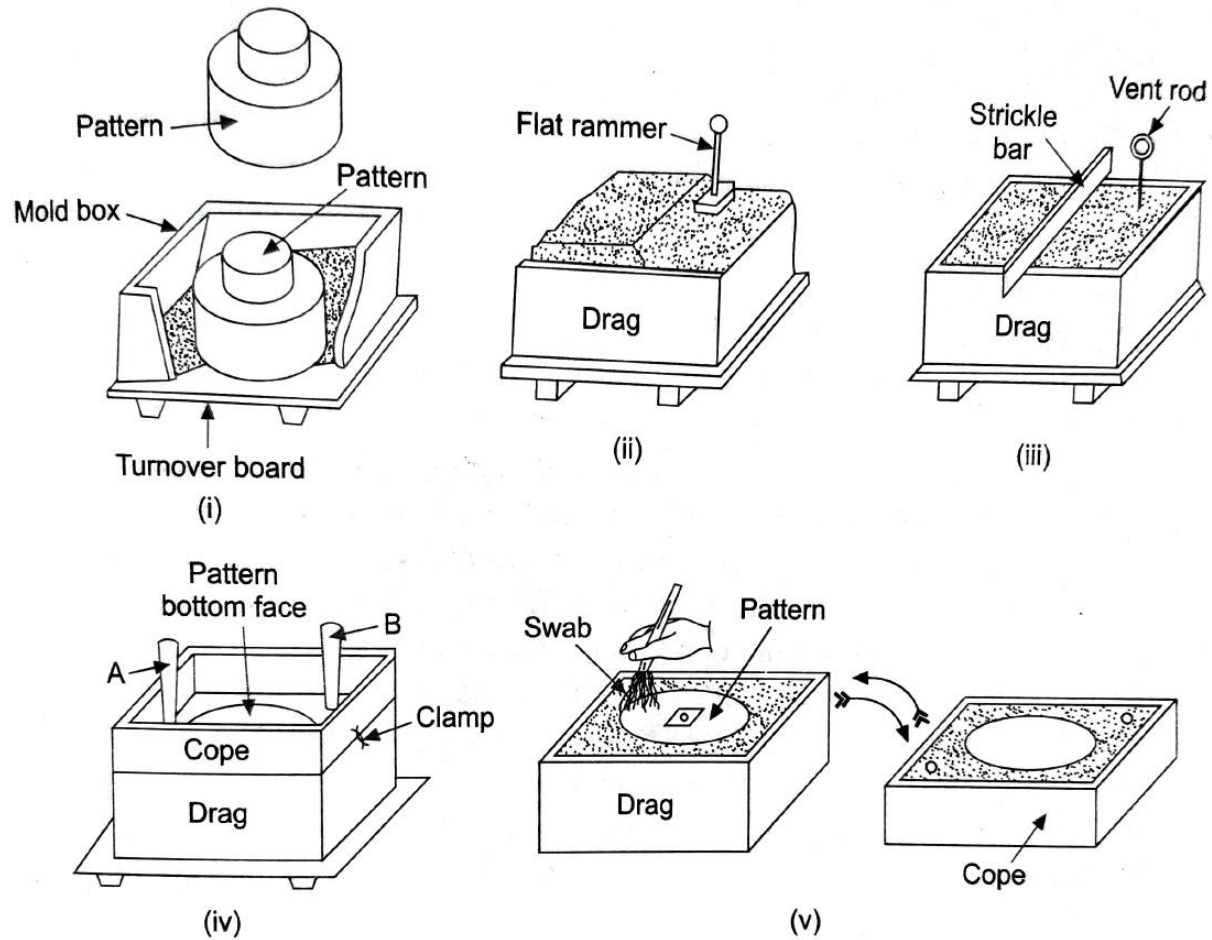
Various casting processes (discussed later) need different types of molds which may be broadly categorized as follows.

- (a) **Expendable molds** such as sand mold, plaster mold, shell mold, investment casting mold, etc., which are used only once (for casting) as these have to be broken up for removing the casting from them.
- (b) **Permanent molds** are made from steel or graphite and are so designed that they can be used repeatedly for a number of times to produce a large number of castings (as the casting can be removed from the mold without breaking it). Examples of such molds include: molds for die-casting, continuous casting and centrifugal casting.
- (c) **Composite molds** are made of two or more different materials, such as sand and graphite molds are used for casting aluminium alloy torque convertors.

### **Salient features of a good sand mold**

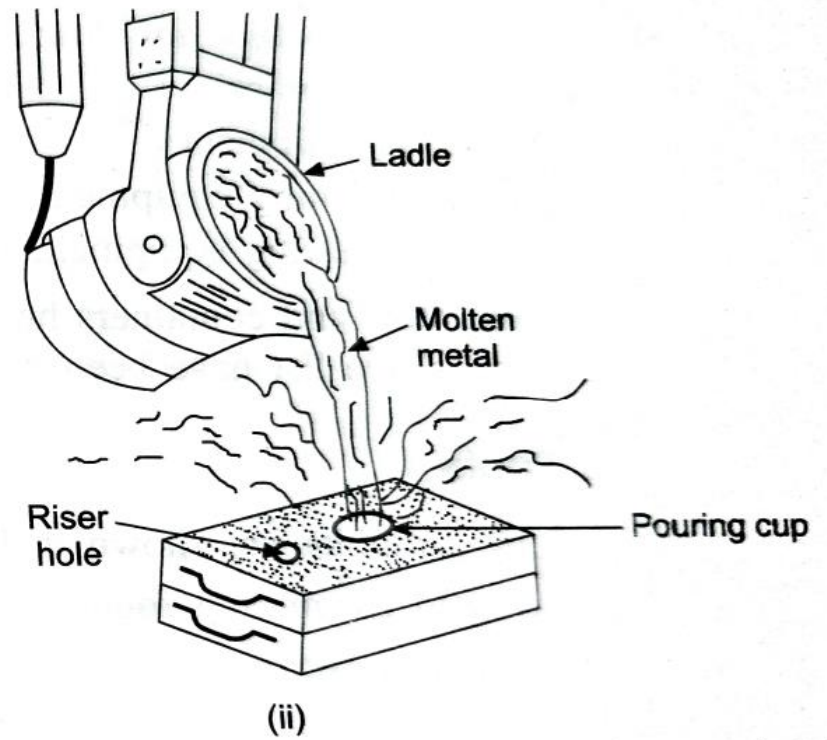
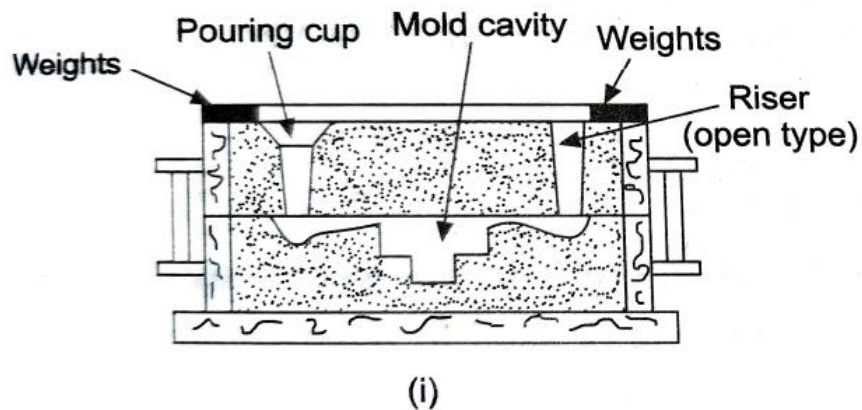
A good sand mold should possess the following specific properties to ensure production of sound castings.

- (i) Properly designed gating system to provide adequate and turbulence free flow of metal in the mold.
- (ii) Generate minimum amount of gases on interaction of molten metal with mold.
- (iii) Good venting capability to vent out gases generated during casting.
- (iv) Good capability of resisting erosive action of hot metal flowing in the mold to ensure correct shape and size of casting.
- (v) Higher refractoriness to withstand high temperatures of molten metal without the fusion of sand.
- (vi) Good strength to resist weight of molten metal.
- (vii) Collapsibility for giving way to free contraction of solidifying casting for avoiding hot tears and cracks in casting.
- (viii) Capability to provide specific rate of cooling for the casting so that a casting of desired structure may be produced.



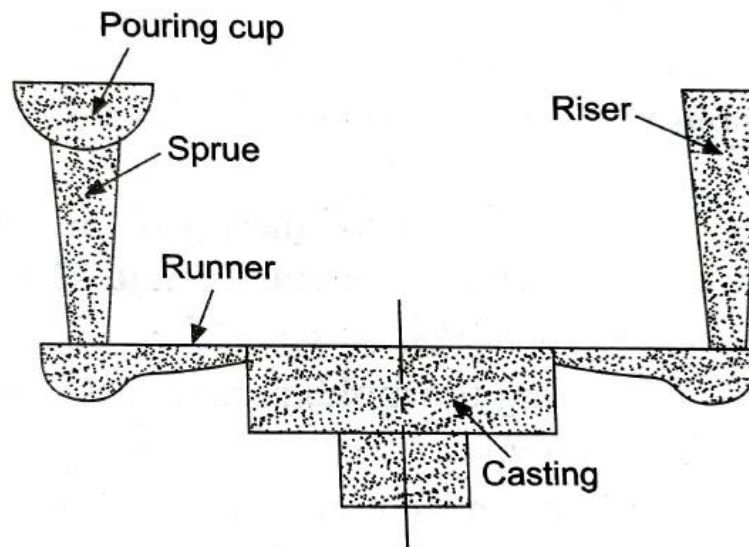
**Fig. 5.2(a)** Steps involved in molding a simple block.





**Fig. 5.2(b)** Sectional view showing (weights not shown). The state of mold just before pouring metal at (i) and metal pouring in the mold at (ii).





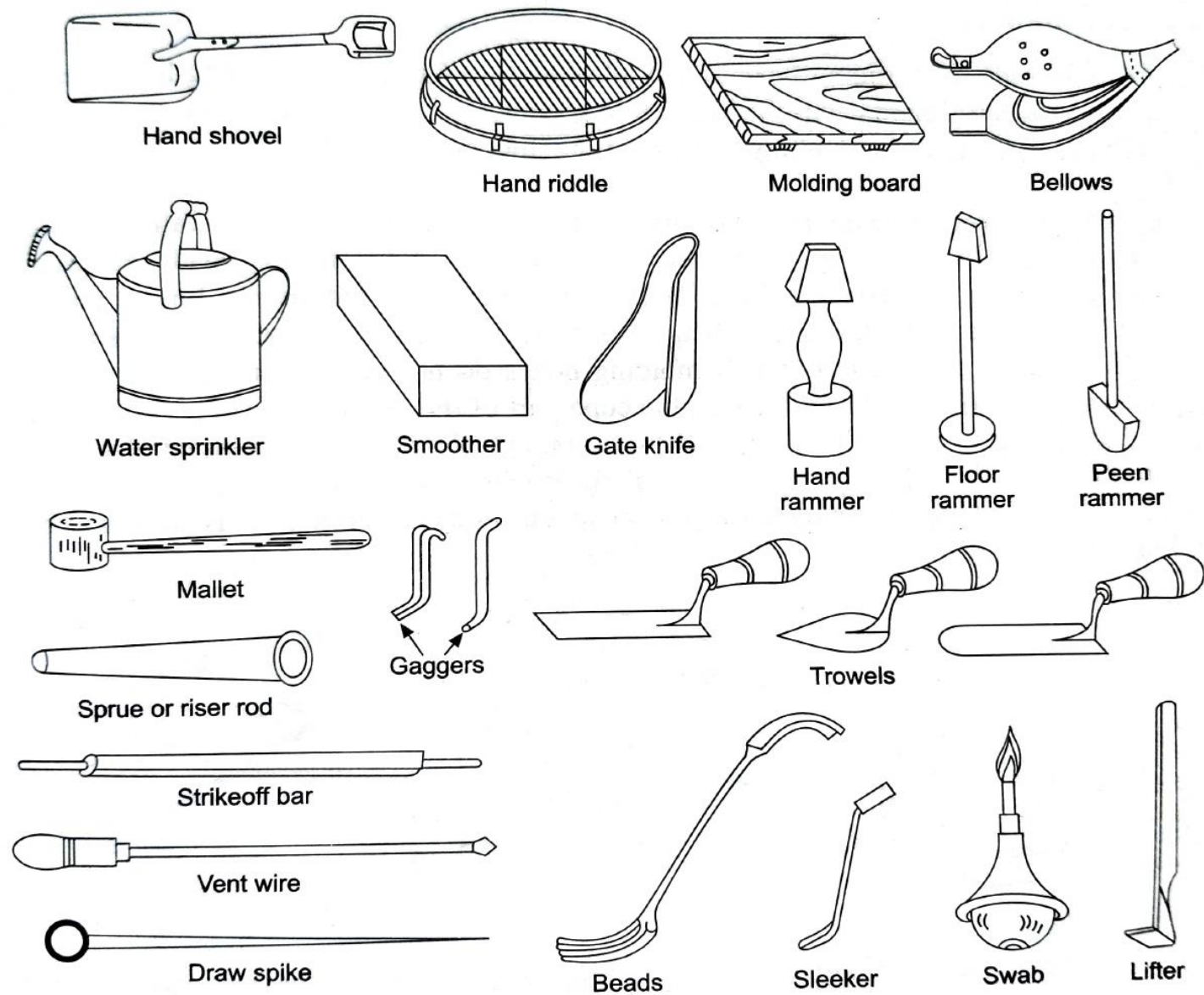
**Fig. 5.2(c)** Shape of casting as it leaves the mold box. The cast portion of riser, pouring cup, sprue and runner are removed to get the actual casting.

### 5.7.1 Hand Tools

Hand tools used by a molder are shown in Fig. 5.3.

- **Shovel** is used for mixing foundry sand and filling it in the mold boxes.
- **Hand riddle** is a sorting device used for riddling of sand to remove foreign matter (like nails and wires) when sand is sieved through it.
- **Water sprinkler** is a handy device for wetting and tempering the sand.
- **Rammers** are used for packing and compacting molding sand floor or molding sand in the molding box. Different types may include: floor rammer, hand rammer and peen rammer.
- **Molding board** or **turnover board** is used either for bench molding or for making a small mold on foundry floor. Wooden boards are quite common.
- **Strickle bar** or **strikeoff bar** is a flat bar of wood or metal and is used for striking off the excess sand from the mold box after ramming. It has one edge bevelled with the surface perfectly smooth.
- **Vent wire** is a thin steel rod with a pointed edge and is used to prick the rammed sand filled in the mold box for making small holes or vents for the escape of gases and steam during casting.

- **Trowels** of different shapes are used for the repair or mending the broken portion of the sand mold after the pattern is taken out from it. These are also used sometimes for filling sand in small mold boxes.
- **Slicks** are also used for repairing the mold, particularly the external or internal round and square corners of the mold.
- **Brush** usually made from cotton is used for clearing the parting sand from the surface of the pattern.
- **Swab** has small bulk of fibres and is used to dampen the outer edges of the pattern before it is taken out from the mold or compacted sand.
- **Cleaners** or **lifters** are finishing tools and used for repairing the mold in deep places after the withdrawal of pattern from the mold. They are also used to remove loose sand from the mold cavity.
- **Beads** are also used for repairing the mold.
- **Draw spike** or **screw** is a steel rod with a loop at one end and pointed or screwed portion on the other end. It is used to rap and lift or pull out the pattern from the mold.



**Fig. 5.3** Hand tools used by a molder.



- **Smoother** is a rectangular wooden block used to smoothen the upper surface of the mold box filled with sand. Metallic smoothers are also used.
- **Mallet** is a wooden hammer used to drive draw spikes in the pattern for pulling it out of mold (or sand).
- **Gate knife** is used to make pouring basin or cup by enlarging and giving an appropriate shape to the metal receiving end of the pouring sprue in the mold.
- **Sprue rod** and **riser rod** are tapered cylindrical wooden rods used for making metal pouring sprue and riser hole in the cope (upper half of the mold).
- **Bellows** are used to blow out dust and sand particles from the mold cavity and thus clean it by air pressure.
- **Shake bag** is used to sprinkle parting compound on the mold and is made of cotton clothe.
- **Gaggers** are bent pieces of wires and rods used for reinforcing the downward projecting sand mass in a cope (or top portion of mold) in making large size molds.
- **Nails** and **wire pieces** are used to reinforce cores or thin projections of the sand in the mold.



# Moulding Sands

- MOLDING SANDS (OR FOUNDRY sand) mixtures or molding mixtures **used for making molds** in a foundry shop.
- Different types of molds such as green-sand mold, skin-dry sand mold, dry-sand mold, CO<sub>2</sub> hardened sand mold, etc. are all made from foundry sands of various types and grades
- Molding sands are broadly **classified according to the nature of their origins.**

# Natural Sands

- Natural sands are **collected from natural resources** like rivers, lakes seas or deserts.
- Natural sands **constitute the major part** of all foundry sands.
- They **contain silica sand, clay substances and water**.
- Water plays an important role in molding sands. When present up to 5 to 8%, water **imparts green strength to the sand in the presence of clay**, besides improving **workability and permeability**. Because of being moist, natural sand is also called **green sand**.

# Synthetic Sands

- Synthetic sands are **prepared from silica sands** which are **relatively clay free**.
- They **need some binder** to make them suitable for foundry work.
- When **mixed with suitable binders and additives**, the silica sands are **converted into synthetic sands**.
- These are considered better sands (than natural sands) as their **properties can be easily controlled** by varying the contents of sand mixture.
- None of the natural sands possesses the required qualities to the extent necessary for being a good foundry sand, since these sands lack one or or more importantly required properties which are always covered up by blending or mixing other sands or materials (like additives) with the natural sand, so that the resulting sand mixture turns out to be a good sand for foundry purposes.

# Special Sands

- In addition to the natural sands and synthetic sands which are primarily based on silica sands here are certain other varieties of special sands such as **zirconite and olivine** which are **better in performance** than silica sand but being **costlier**, these are used only for special applications.

# CONSTITUENTS OF MOLDING SANDS

The principal constituents of a molding sand are:

- Silica sand
- Binders
- Additives
- Water



# Silica sand

- It **forms the bulk** of the foundry sands and imparts **refractoriness** and other properties to the molding sand.

# Binders

- Binders give **cohesiveness and strength** to the sand to enable it to **retain the shape** of the mold cavity after the withdrawal of the pattern from the sand.
- Binders may be **organic** type (such as linseed oil, molasses, dextrin, pitch or resins) and **inorganic** type (such as clays or cement).
- Clay binders include: bentonite, limonite, fire clay, etc.
- Clay binders are used for both green-and dry-sand molding.
- Increased clay content adds to **strength, hardness and toughness** of the sand but **reduces flowability** and **permeability** of sand.

# Additives

- These are used to **improve the properties of the sand**, either by improving the existing properties of the sand or by imparting new properties to make the sand mixture more useful.
- Common additives are:
  - coal dust, sea coal, cereals, silica flour, pitch, dextrin, molasses, wood flour, etc.

# Additives

- **Coal dust**, when present (up to 10%) in the molding sand, **reacts with oxygen present** in sand pores at the time of casting and produces a reducing atmosphere of CO<sub>2</sub> at the mold- metal interface, which **gives smooth castings**.
- **Cereal flour** improves **strength and collapsibility of sand** (to allow free contraction of casting).
- **Silica flour** increases **hot strength of mold** and **decreases metal penetration into mold walls** and thus gives smooth castings

# Additives

- Both **dextrine** and **molasses** improve **dry strength** (after baking the mold) and **decrease metal penetration into mold walls**.
- Additives such as **wood flour**, **cereals** and **cellulose** when added to molding sand work as **cushion materials** since they **burn and form gases** during casting, thereby giving rise to **space to accommodate expansion** of casting during its cooling.

# Water

- **Adding water** to molding sand is called **tempering**.
- Water imparts **workability** to the sand for making the mold.
- Binders, such as clays give green strength to the sand **due to the presence of water**.
- A **green-sand** mold may contain water around **5%** whereas the sand for making a **dry-sand mold** (which is later baked) has higher water contents, up **to 10%**.
- **Too high water causes blow holes** in the castings.
- **Low moisture** sands are **weak in green strength** and produce defects such as scabs or roughness on the surface of castings.



# SAND PREPARATION AND CONDITIONING

- None of the naturally available sands possesses the desired the required degree for being considered a good molding sand.
- By the process of **blending or mixing varieties of sands together** and by **adding binding materials or additives** to this mixture, a new **suitable sand mixture is produced** for foundry purposes.

# SAND PREPARATION AND CONDITIONING

- New sands as well as used floor sand (which has been used several times in making molds) are properly prepared and mixed in a suitable ratio.
- For example, the **ratio** of used floor sand (old) to new sand may be **more for light castings** but it **decreases for medium and heavy castings**

# SAND PREPARATION AND CONDITIONING

- Proper conditioning means **uniform distribution of clay bond and other additives over sand grains**, even **distribution of proper moisture** and sorting out of **foreign matter like nails, and other metal pieces** (which might have been used for strengthening previous mold walls) by riddling and a thorough mixing of sand mass.
- The operation of conditioning is carried out **manually by mixing the sand mixture and other additives with hand shovels**.
- **Modern foundries** manually have appropriate equipment for conditioning of the sand such as **circular pan sand mixer** using rotating stone wheels and paddles.
- **Testing of sands** is also carried out for strength, permeability, moisture content, etc. to predict its performance during use.

# Properties of Moulding Sand

## 1. Refractoriness

- property which makes the molding sand capable of withstanding high temperatures of molten metal without the fusion of the sand.
- Silica sand present in the molding sand is primarily responsible for imparting this property
- higher contents of impurities such as lime magnesia, metallic oxide, etc. present in sands tend to lower the fusion point of silica sand
- Larger grains of silica increase refractoriness.

## 2. Permeability or porosity

- property of sand which **allows easy escape of gases** and steam through the sand mold when molten metal comes in contact with moist sand having coal dust, oils, resins and other gas forming agents
- Most molten metals also **have dissolved gases in them** which are evolved on solidification
- Insufficient porosity of molding sand leads to several casting defects such as porosity, honeycombing etc.
- Permeability **depends** on the **shape and size of sand grains, the clay, moisture contents** and **ramming** of sand (around the pattern during molding) and **venting of the mold**

### 3. Cohesiveness

- ability of sand particles to stick together and thus gives green strength to the moist molding sand in maintaining the shape of mold (or mold cavity) after the pattern has been withdrawn from the sand.
- Depends on the size, shape and distribution of sand grains, type and contents of clay and other bonding materials and the moisture contents

## 4. Plasticity

- property of a molding sand by virtue of which it **takes easily any desired shape** (as per the pattern) **under pressure** and **retains it** after the pressure is removed.
- **Fine-grained sands** give good plasticity
- Plasticity also depends On the **clay contents**, which absorb moisture and help improving the plasticity.



## 5. Flowability

- property due to which the sand flows under the effect of ramming (during molding) to all portions or corners of the molding box and packs properly around the pattern while distributing the ramming pressure evenly on the sand in all parts of the molding box.

## 6. Adhesiveness

- provides the sand the capability of easily adhering to the surfaces of other materials such as the walls of molding boxes
- helps in retaining the sand (filled in the molding box) intact during molding (when mold boxes filled with sand may be lifted or overturned during the process)

## 7. Collapsibility

- Property because of which a sand mold (or core) collapses automatically, giving way to free contraction of the solidifying casting and thus avoids cracks in castings

## 8. Other properties

- low coefficient of expansion;
- property of non-sticking with casting;
- non-reacting chemically with molten metal;
- low cost
- easy availability

# Some commonly used Sand Mixtures (Types of Molding Sands)

## 1. Green sand (tempered sand):

- **moist molding sand** usually available in foundry shops for **general molding purposes**
- Molds made with this sand called **green-sand mold**
- molten metal is **poured** during casting **without any Prior drying or baking** of the mold
- made by **mixing natural sand and other sands** (including used floor sand) and **additives** and contains **just enough moisture** (up to 8%) to give sufficient bonding strength to the mold
- **least expensive** foundry sand and **takes less time in making** the mold ready for casting (as **no drying or baking of mold is involved**).
- **Small and medium sized castings** of ferrous and nonferrous metals are made in this sand

## 2. Dry sand

- molding sand which at the time of making a mold has excess moisture but the same has been evaporated by drying the internal mold face or the full mold in an Oven.
- The mold thus made is called dry-sand mold.
- sands used for these molds are fine- grained and mixed with proper binders and additives that give strength to the mold on baking
- used for casting large size steel and cast iron components.

### 3. Floor sand / Black sand / Backing sand

- molding sand which is **used over and over again** for molding purposes
- usually **black in colour** due to the **addition of coal dust** and also **due to burning as result of coming in contact with the molten metal repeatedly** during casting
- It usually **forms the bulk of the molding sand** as it **supports the facing sand** filled only to a certain depth around the pattern during molding.
- mixed in **certain proportion** with the **natural sand, clay or other binders and additives** while preparing the new green-sand molding mixture.

## 4. Facing sand

- fine textured sand used to form the face of the mold cavity
- used around the pattern to a thickness of about 5 cm
- certain amount of floor sand and new molding sand properly tempered with moisture and suitable additives give a good facing sand.
- facing sand has better strength and high refractoriness than the floor sand.



## 5. Parting sand

- usually a natural dry silica sand sprinkled on to the pattern and also on the parting surfaces of the mold (when made in two or more boxes), so that the sand mass of the molding boxes may not stick to each other or to the pattern
- parting compound made from phosphate rock is also used sometimes known as parting sand

## 6. Core sand (or oil sand)

- large grain high silica sand bonded with organic binders such as linseed oil, light mineral oil, corn flour, etc.
- It finds use in making cores and molds for non-ferrous castings

## 7. Loam sand

- comprises a mixture of ordinary clay (about 50%), fire clay and silica sand and water milled to a thin paste which, in the form of plaster, is applied on the face of a roughly carved mold made by using burnt clay bricks.
- The mold is brought to the final shape with a rotating sweeping pattern which molds the plastered face to the required shape and size of the mold
- The plaster hardens on drying
- symmetrical objects (cylinder, barrels, kettles) are made in such molds

# Patterns

- used for **molding a cavity** (or mold) in the molding sand mixture such that the **formed cavity is similar to the shape of the casting**
- The pattern is a **replica of the desired article to be cast** but **differs from the actual article** in certain ways:
- pattern carries
  - (a) **additional allowance in its dimensions** to compensate for the **metal shrinkage** during casting;
  - (b) **allowances for machining or finishing** the castings;
  - (c) **draft or taper** on its exterior and interior surfaces for its **easy removal** from the molding sand;
  - (d) **additional projections (core prints) to produce seats** in the mold for the setting of cores.

# Patterns

- Patterns are made from **wood, metals, rubber, plaster, wax, plastics**, etc.
- The selection of a particular material for making pattern depends on factors such as:
  - **number of castings** to be made,
  - **method of molding** (hand or machine),
  - **quality of castings** and degree of finish and dimensional accuracy desired on castings,
  - design of the casting with possibilities of **changes in the design of the pattern** and
  - expectations of **repeat orders**

# Pattern Colours

- **Yellow**- Core Prints
- **Red**- Surfaces (of the resulting casting) to be machined
- **Black**-Surfaces (of the resulting casting) to be left unmachined
- **Red strips on yellow base**- Seat for loose pieces of the pattern
- **Clear or no colour**-Parting surface

# MATERIALS FOR PATTERNS

## 1. Wood:

- commonly used because it is cheap, easily available, can be shaped easily, light in weight and can have good surface finish.
- however, is affected by moisture and changes its shape and size on drying out.
- It also wears out quickly against abrasion by molding sands and hence has comparatively shorter life
- wood patterns are not used for the production of large quantities of castings
- The wood used for patterns include white pine, deodar, tun, walnut, teak, etc

## 2. Metals

- Metals used for making patterns include cast iron, brass, white metal, aluminium etc.
- Metallic patterns are preferred when a very large number of castings are to be made
- costlier than wooden patterns but have much longer life.



### 3. Plaster of Paris and gypsum cement

- Plaster of Paris and gypsum cement are **quick setting compounds** used for making patterns.
- They are **easily cast into intricate shapes**.
- **Small patterns of complicated shapes** are made of plaster of Paris by pouring and setting the plaster slurry into the pattern molds.

## 4. Plastics

- Plastic patterns are **lighter, stronger and dimensionally more stable** than wooden patterns.
- Pattern surface being **smooth**, molding sand **sticks less to the pattern**.
- **no moisture absorption** by the pattern
- Wooden patterns are, however, **easy and quick to make and repair at cheaper cost**
- Among thermoplastics, **polystyrene** is commonly used for patterns (known as **consumable pattern** as heat of molten metal vaporizes the pattern which leaves behind the formed mold cavity).
- Thermosetting plastics such as **phenolics and epoxies** are also used for patterns

## 5. Wax

- used in investment casting process (or lost wax process).

# PATTERN ALLOWANCES

## 1. Shrinkage allowance

- Most metals, when they cool down from molten state to room temperature, **shrink or contract** in three different stages:
  - (i) **Liquid contraction**,
  - (ii) **Solidifying contraction** and
  - (ii) **Solid contraction**.
- The **first two types** of contractions (involving change from liquid to solid) are **compensated by the feeding of metal from risers** where the metal remains in molten state till the casting gets solidified.
- It is for the **third type** of contraction, the solid contraction (contraction of solidified casting to room temperature), for which **allowance is provided on the patterns**.

# 1. Shrinkage allowance

- Different metals have different contraction allowance.
- The solid contraction of the metal is influenced by:
  - the metal to be cast
  - the pouring temperature of the molten metal,
  - design, geometry and size of casting:
  - type of mold material and method of molding
  - resistance of mold to shrinkage

# 1. Shrinkage allowance

- Although the contraction of the castings is volumetric contraction, the **contraction allowance is, however given in linear measures**

**TABLE 5.1** Solid contraction allowance for different metals

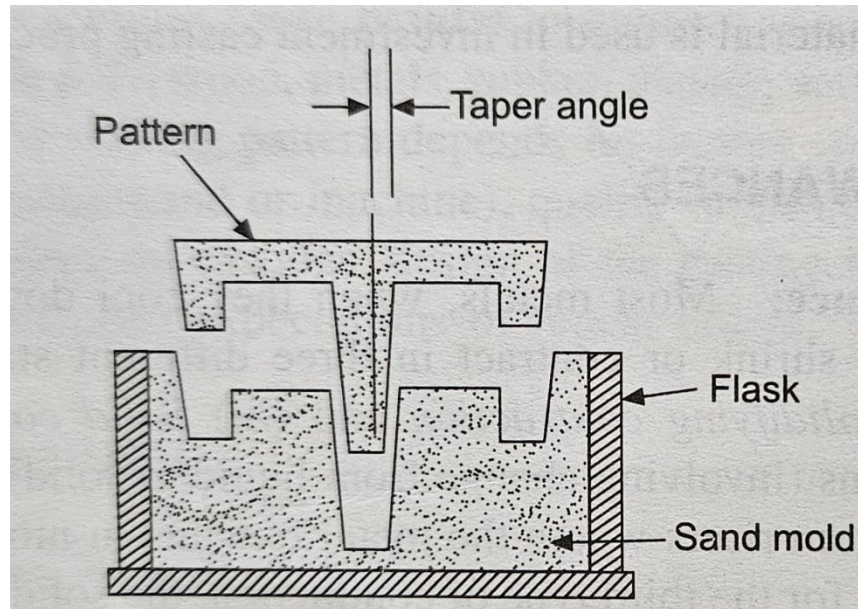
| <i>Metal to be cast</i> | <i>Allowance (mm/metre)</i> |
|-------------------------|-----------------------------|
| Grey cast iron          | 10.5                        |
| Malleable cast iron     | 10.5                        |
| Steel                   | 21.0                        |
| Brass                   | 16.0                        |
| Aluminium               | 16.0                        |
| Zinc                    | 24.0                        |
| Lead                    | 24.0                        |
| Copper, magnesium       | 16.0                        |
| Silver                  | 10.0                        |
| White cast iron         | 21.5                        |

## 2. Machining allowance

- Given on a pattern only when the casting being made **needs machining and finishing**
- This allowance is given **in addition to the shrinkage allowance** and varies **from 2 to 5 mm** according to the location and nature of casting, method of machining and the degree of finish required.
- In **larger castings** the machining allowance may be **12.5 mm or more**

### 3. Draft allowance

- Patterns are given slight taper on their vertical surfaces (both external and internal) which are parallel to the direction of their withdrawal from the mold





### 3. Draft allowance

- This taper or draft is expressed either in degrees or in terms of linear measurements, amount of draft being more on internal surfaces.
- The draft amount may vary from 10 to 25 mm per metre on external surfaces and from 40 to 70 mm per metre on internal surfaces.

## 4. Rapping or shake allowance

- To take the pattern out of the compacted sand mold the pattern is first rapped or shaken by striking over it side to side, so that the pattern surfaces may be made free of the adjoining sand walls of the mold.
- This operation obviously enlarges the size of the mold cavity and hence a negative rapping allowance is provided on patterns to compensate, at least in case of large castings